Geographic and Seasonal Analysis of Surface Water Quality in India Using CPCB Monitoring Data

Abstract

Surface-water quality is a major environmental concern in India due to the diverse range of rivers and other surface-water resources and the extensive monitoring carried out across the country. This project proposes an exploratory and statistical analysis of surface-water quality in India using official monitoring data obtained from the National Water Data Portal (NWDP) and the Central Pollution Control Board (CPCB). The primary dataset consists of 315 state-wise CSV files containing more than 300,000 monitoring observations covering physical, chemical, and biological water-quality parameters. The analysis will focus mainly on Dissolved Oxygen (DO), Biochemical Oxygen Demand (BOD), pH, and Electrical Conductivity, along with geographic attributes such as state, district, monitoring station, latitude, and longitude. A supplementary Indian Water Quality Dataset obtained from Kaggle, containing approximately 1,991 observations and 12 variables for the period 2003–2014, will be used to provide historical context. The study will investigate how DO, BOD, and pH vary across Indian states and geographic regions; how DO and BOD vary across pre-monsoon, monsoon, post-monsoon, and winter seasons; what relationships exist among DO, BOD, pH, and Electrical Conductivity; and how state-level DO and BOD measurements from 2003–2014 compare descriptively with the CPCB monitoring period of 2018–2021. The analysis will involve data loading, cleaning, missing-value assessment, descriptive statistics, correlation analysis, and visualization using base R and `ggplot2`. Bar charts, scatter plots, boxplots, distribution plots, and geographic visualizations will be used to identify patterns and variations in surface-water quality. The historical dataset will be analyzed separately and compared with aggregated CPCB results rather than directly merged. The project is expected to demonstrate the usefulness of large-scale environmental monitoring data for understanding geographic and seasonal patterns in Indian surface-water quality.